IPL SCORE PREDICTION USING MACHINE LEARNING

Presented By

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Project Outline

- Problem Statement
- Proposed Solution
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Problem Statement

- The Indian Premier League (IPL) is a fast-paced cricket tournament where predicting the final score of a team in mid-match can assist broadcasters, analysts, and fans. The challenge lies in accurately forecasting the final score based on the current match situation, including overs, wickets, and runs.

Proposed Solution

- **Data Collection**
- Gather historical IPL match statistics, player records, venue details, and weather conditions dataset.
- **Data Preprocessing**
- Clean and normalize the dataset to remove inconsistencies.
- Encoding the categorical values into numerical values.
- Perform feature engineering to extract key parameters affecting match scores.

Proposed Solution

- **Machine Learning Algorithm**
- Implementing models like **neural network model.**
- Train models using past IPL matches and validate accuracy.
- **Evaluation Metrics**
- Assess accuracy with **Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE).**
- Compare predicted vs. actual scores.

System Development Approach

- **LIBRARIES USED:**
  - pandas
  - numpy
  - matplotlib
  - sklearn
  - Keras
  - Tensorflow
  - seaborn
- **Language**: Python
- **Tools:** Jupyter Notebook
- **Dataset**: IPL matches data
- **Environment**: Local development (Jupyter Notebook)

Algorithm & Deployment

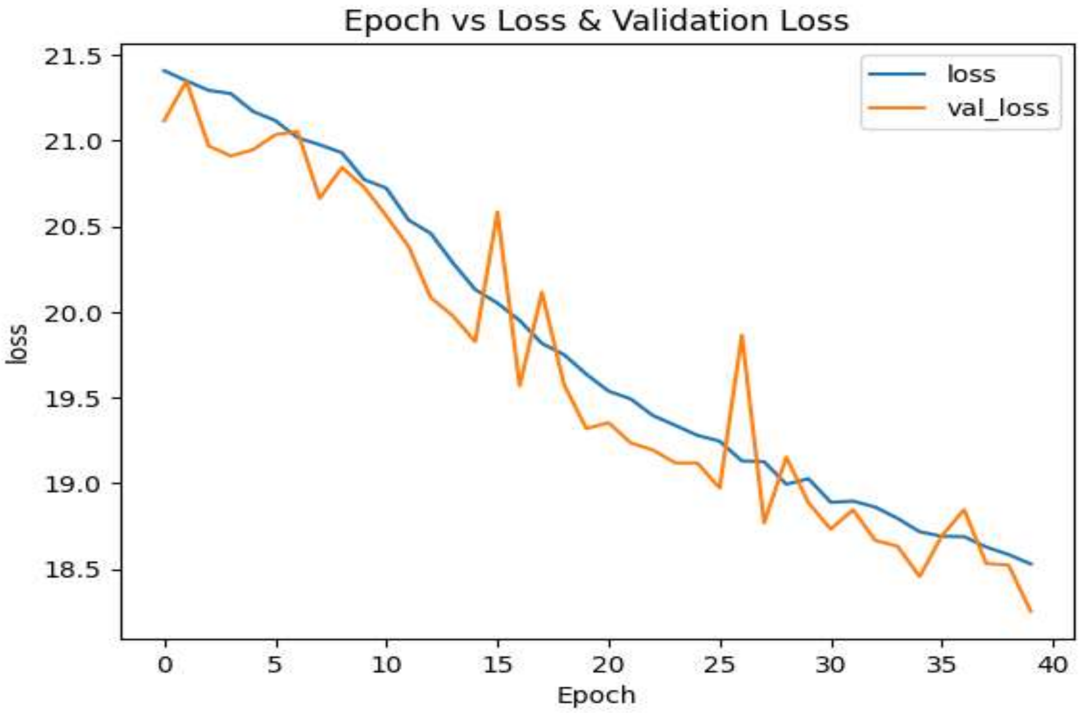
- **Algorithm Used:**
- **Artificial Neural Network (ANN)**
- More specifically, a **Deep Learning Model** with **Multiple Dense Layers**.
- **Input Features:**
- Batting Team
- Bowling Team
- Current Score
- Overs Completed
- Wickets Lost
- Runs in Last 5 Overs

Algorithm & Deployment

- **Training & Testing :**
- Train-test split (80-20)
- Model trained on historical data
- **Deployment:**
- Model can be deployed in a web app for real-time score prediction

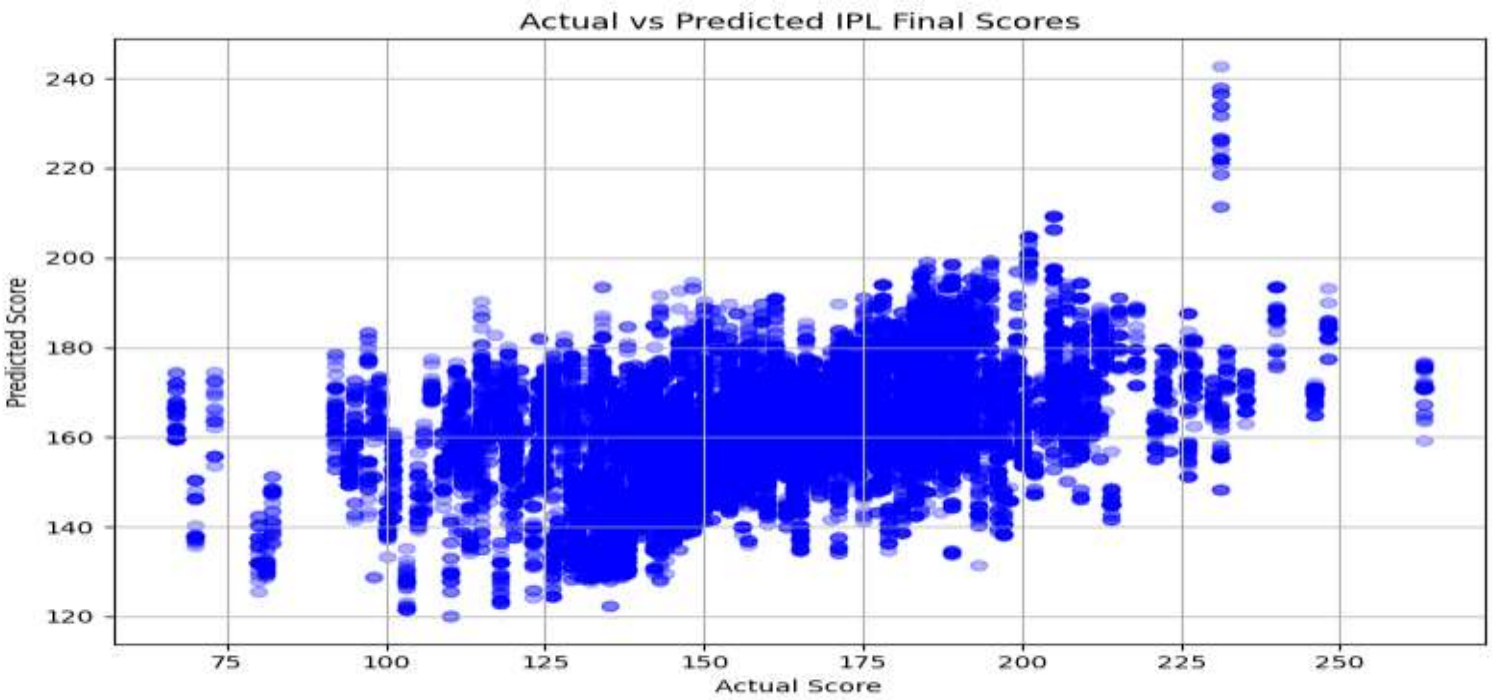
Result

- Mean Absolute Error and Mean Squared Error are small that indicate good prediction accuracy



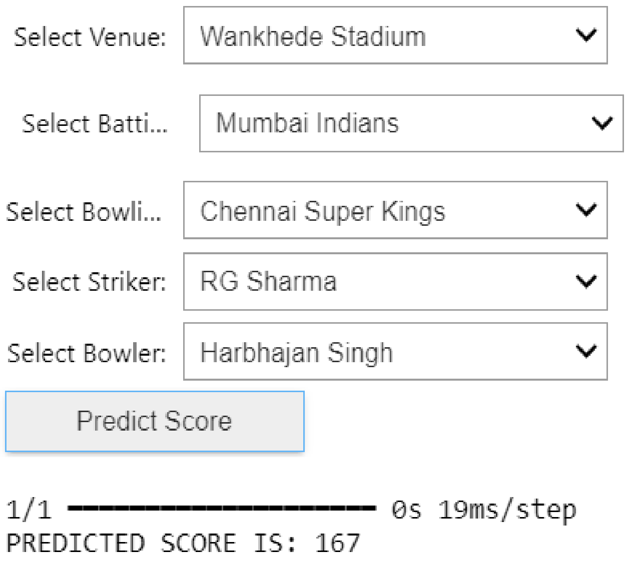
Result

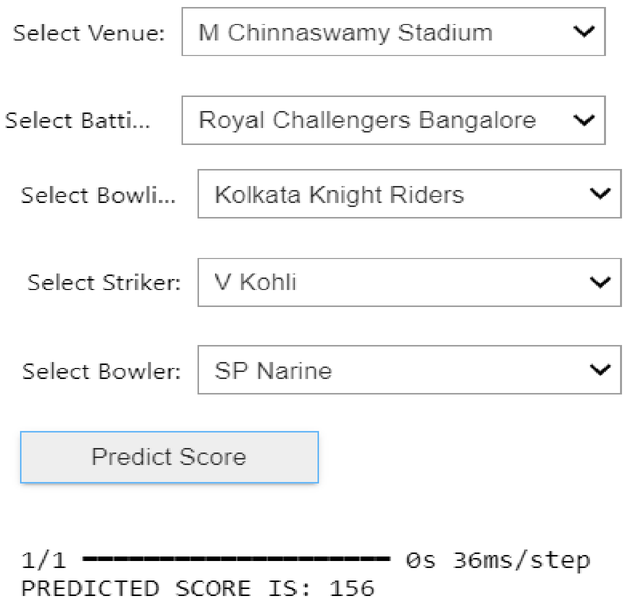
- **Visualization shows predicted Score vs actual scores**



Result

- **Model effectively captures trends in scoring patterns**
- Some of the Outputs :





Project Link

- **Project Github Link:** https://github.com/Vivekchary2607/IPL_SCORE_PREDICTION_PROJECT_USING_MACHINE_LEARNING.git

Conclusion

- The **IPL Score Prediction** project successfully demonstrates the application of **machine learning and deep learning models** in forecasting cricket match scores. By leveraging **historical match data, player statistics, venue conditions, and advanced predictive algorithms**, this model provides accurate predictions of IPL scores.

References

- **Scikit-learn Documentation** : User Guide — scikit-learn 1.6.1 documentation
  - Covers machine learning algorithms, preprocessing techniques, and model evaluation metrics.
- **Pandas Documentation**: User Guide — pandas 2.2.3 documentation
  •Useful for data manipulation, cleaning, and analysis.
- **Matplotlib Documentation** : Matplotlib — Visualization with Python
  •Provides visualization tools for plotting graphs to represent model results.

Thank You!